Configure IPV6 Addressing

1. **Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).**

Answer:

System Used: Windows

Assign a Static IPv6 Address

Open Command Prompt as Administrator and run:
netsh interface ipv6 set address name="Wi-Fi" address=2001:db8:1::10

Verify the Configuration

Run:
ipconfig

The output should show:
IPv6 Address: 2001:db8:1::10

Result: The static IPv6 address 2001:db8:1::10 has been successfully assigned and verified on the Wi-Fi network adapter.

2. **Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address. Hint: Use the ping6 or ping -6 command for IPv6 testing.**

Answer

System Used: Windows

Assign a Unique Local IPv6 Address
Open Command Prompt as Administrator and run:

netsh interface ipv6 set address name="Wi-Fi" address=fd00:1:1::10

Configure IPv6 Address on Another LAN Device
Example address for the second device:

fd00:1:1::20

Test IPv6 Connectivity
From the first computer, run:

ping -6 fd00:1:1::20

The output should show:

Reply from fd00:1:1::20: time<1ms
Reply from fd00:1:1::20: time<1ms

Result: The unique local IPv6 address fd00:1:1::10 was configured successfully, and connectivity with the LAN device fd00:1:1::20 was verified using ping -6.

3. **Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect. Constraint: Document the steps and the observed address behavior in a text file.**

System Used: Windows

Enable Temporary IPv6 Addresses
Open Command Prompt as Administrator and run:

netsh interface ipv6 set privacy state=enabled

Check the Current IPv6 Address
Run:

Ipconfig

Example output:
Temporary IPv6 Address: 2001:db8:1::abcd

Disconnect and Reconnect the Network
Disconnect the Wi-Fi, reconnect it, and run:

Ipconfig

Check the Temporary IPv6 Address again.

Restart the Computer
After restarting, run:

Ipconfig

Check whether the temporary IPv6 address has changed.

Result: The temporary IPv6 address was configured and checked before and after network reconnection/reboot. The observed address can change because temporary IPv6 addresses are designed to provide privacy.

4. **Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.**

Language Used: Python

Create the Python Script:

```
restaurants = {
   "Pizza Hut": "2001:db8:1::10",
   "Dominos": "2001:db8:1::20",
   "Burger King": "2001:db8:1::30",
   "Subway": "2001:db8:1::40"
}

search = input("Enter restaurant name: ").lower()

found = False

for name, ipv6 in restaurants.items():
   if search in name.lower():
      print("Restaurant:", name)
      print("IPv6 Address:", ipv6)
      found = True

if not found:
   print("Restaurant not found.")
```

Example Output:

```
Enter restaurant name: pizza

Restaurant: Pizza Hut
IPv6 Address: 2001:db8:1::10
```

5. **Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.**

IPv6 Subnetting Scenario and Manual Configuration

AI-Generated IPv6 Subnetting Scenario
Suppose a company has the IPv6 network:

2001:db8:1234::/48

The company needs at least 3 separate subnets. We can use a /64 prefix for each subnet.

Subnet 1: Administration
2001:db8:1234:1::/64

Example host address:
2001:db8:1234:1::10/64

Subnet 2: Development
2001:db8:1234:2::/64

Example host address:
2001:db8:1234:2::10/64

Subnet 3: Testing
2001:db8:1234:3::/64

Example host address:
2001:db8:1234:3::10/64

I selected Subnet 1 (2001:db8:1234:1::/64) for my system.

Configuration Steps on Windows

Open Command Prompt as Administrator.
Find the network adapter name:
netsh interface ipv6 show interfaces
Identify the active network adapter, such as Wi-Fi or Ethernet.

Assign the IPv6 address using:
netsh interface ipv6 add address "Wi-Fi" 2001:db8:1234:1::10

Replace Wi-Fi with the name of the active adapter if necessary.

Verify the configuration using:
Ipconfig

Check that the manually assigned IPv6 address appears under the selected network adapter.

3. Configured Address
IPv6 Address:
2001:db8:1234:1::10/64A

Subnet:
2001:db8:1234:1::/64

4. Result
The IPv6 subnet 2001:db8:1234:1::/64 was selected from the AI-generated scenario, and a valid IPv6 address was manually assigned to the system's network adapter. The configuration was verified using ipconfig.